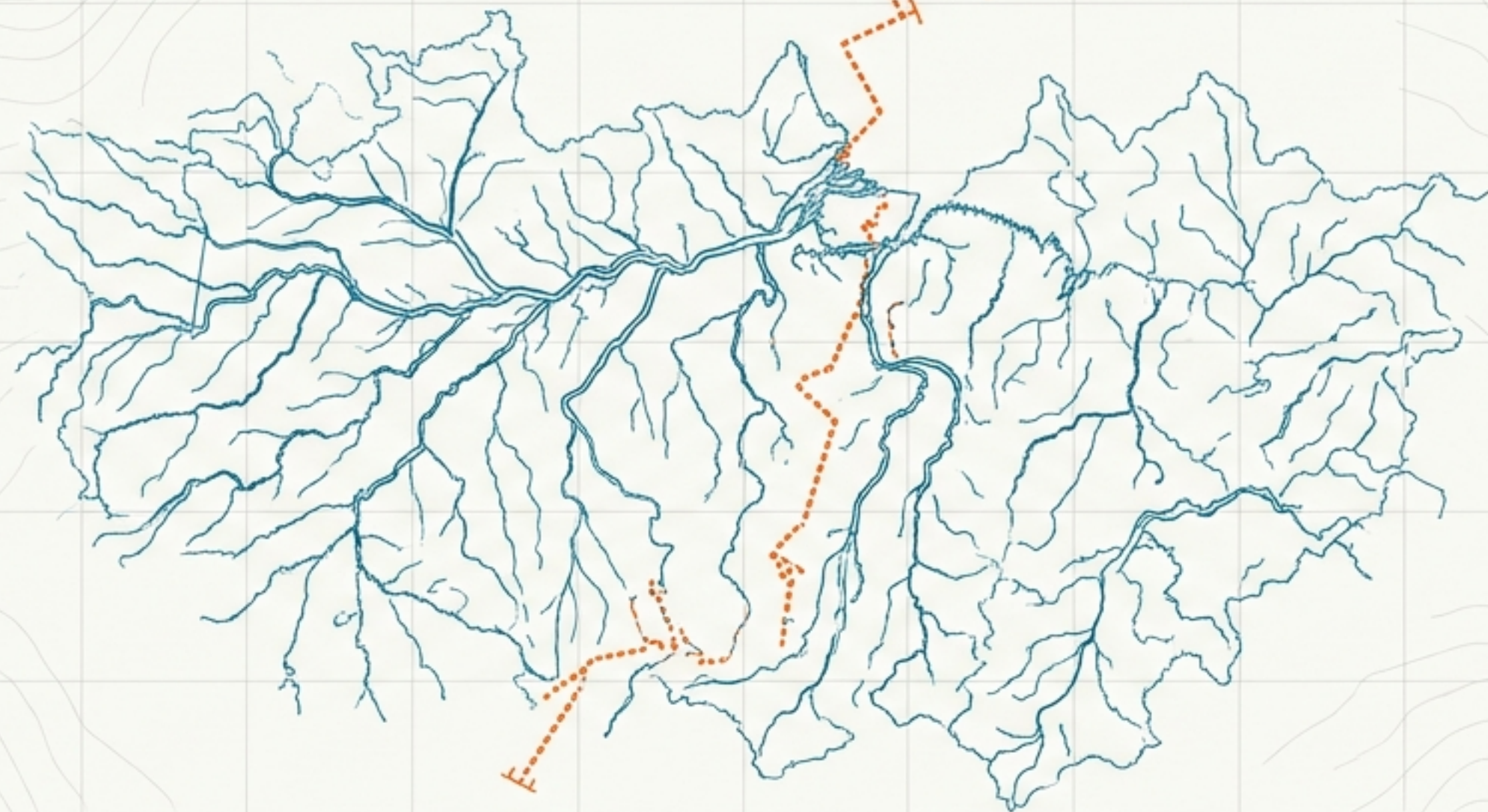




GEOPOLITICAL IMPUNITY & EXTRACTIVE ACTIVITY

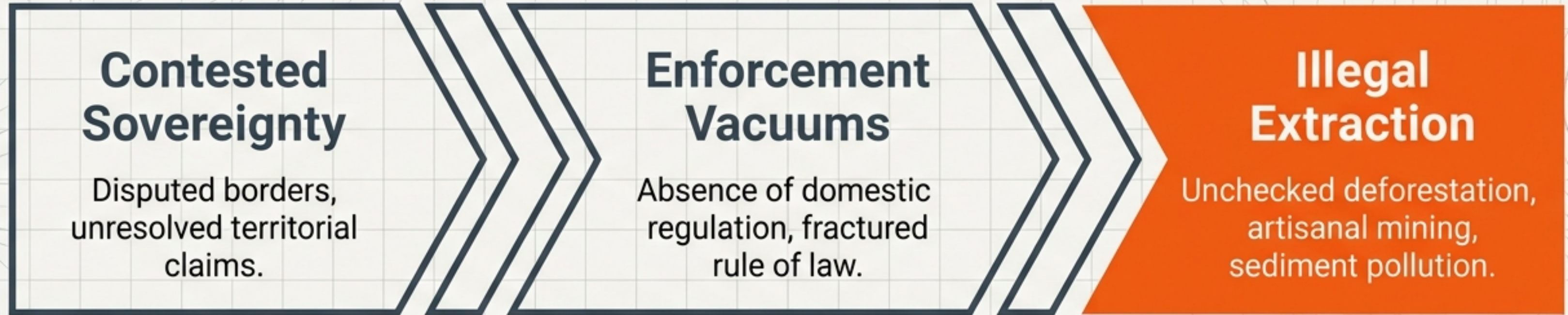
A Hybrid Physics–ML Framework for Detecting Mining Anomalies in Disputed Territories.

[PHD RESEARCH PROPOSAL]

[DATA SCIENCE]

[ENVIRONMENTAL GEOPOLITICS]

The Core Proposition



We can quantify this impunity effect by combining a physical hydrological model with satellite observations. The residual between physical prediction and observed reality is where geopolitics lives.

Bounding the Inquiry

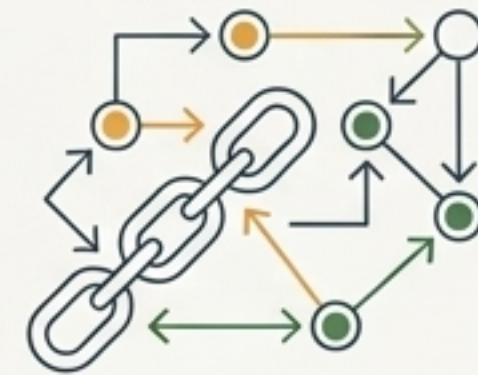
The Primary Question (Detection)

Do territories with active sovereignty disputes exhibit systematic excess extractive activity—such as deforestation, mining, and sediment pollution—**beyond** what natural hydrological and environmental conditions predict?



The Secondary Question (Attribution)

Is the magnitude of that excess statistically predicted by geopolitical features, including dispute status, governance quality, metal prices, and proximity to borders?



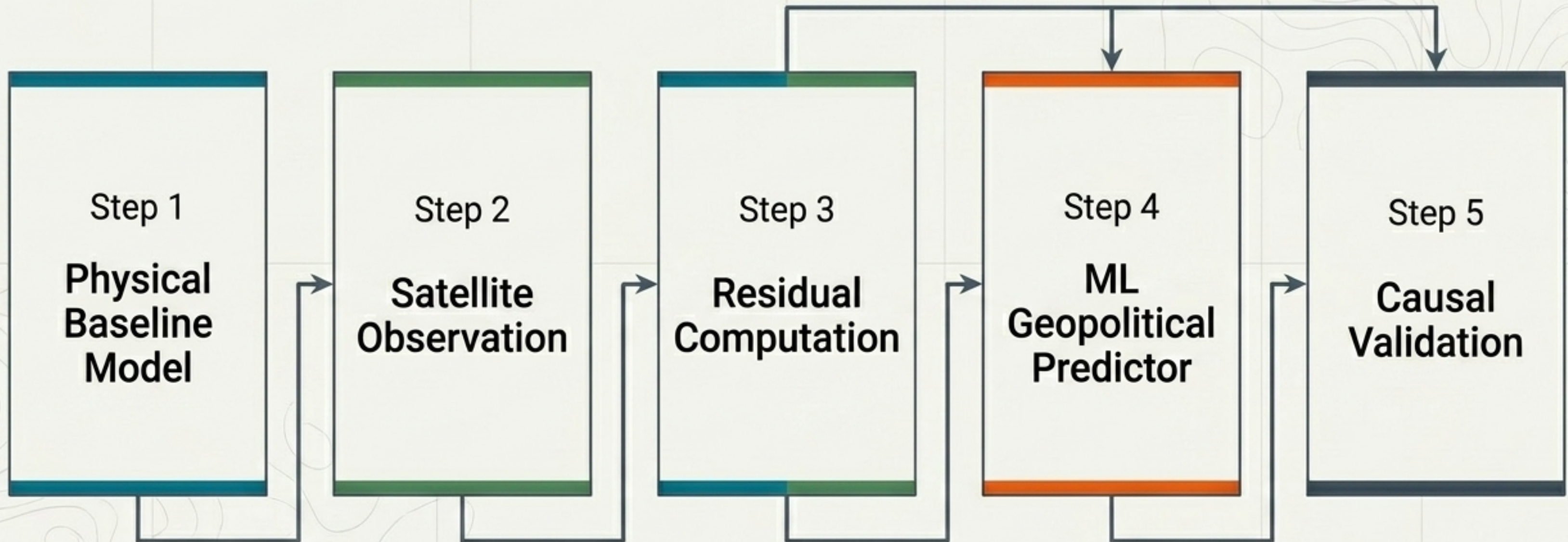
Bridging Isolated Disciplines

	Pure Remote Sensing
Methodology	Pure ML / Change detection.
Resolution	High spatial (pixel-level).
Mechanism	Black-box classification.
Output	Anomaly maps without causal physical baseline.

Pure Geopolitics
Statistical regressions.
Low (Aggregate country-level statistics).
Theoretical correlations.
Broad policy literature.

The Hybrid Proposal
Physics-guided Machine Learning.
Dynamic Catchment-level over time.
Interpretable physical baseline + ML residual correction.
Framed physics-residuals as isolated, quantified geopolitical signals.

Pipeline Architecture: A 5-Step System



Subsequent sections detail the data and computational layers of each pipeline node.

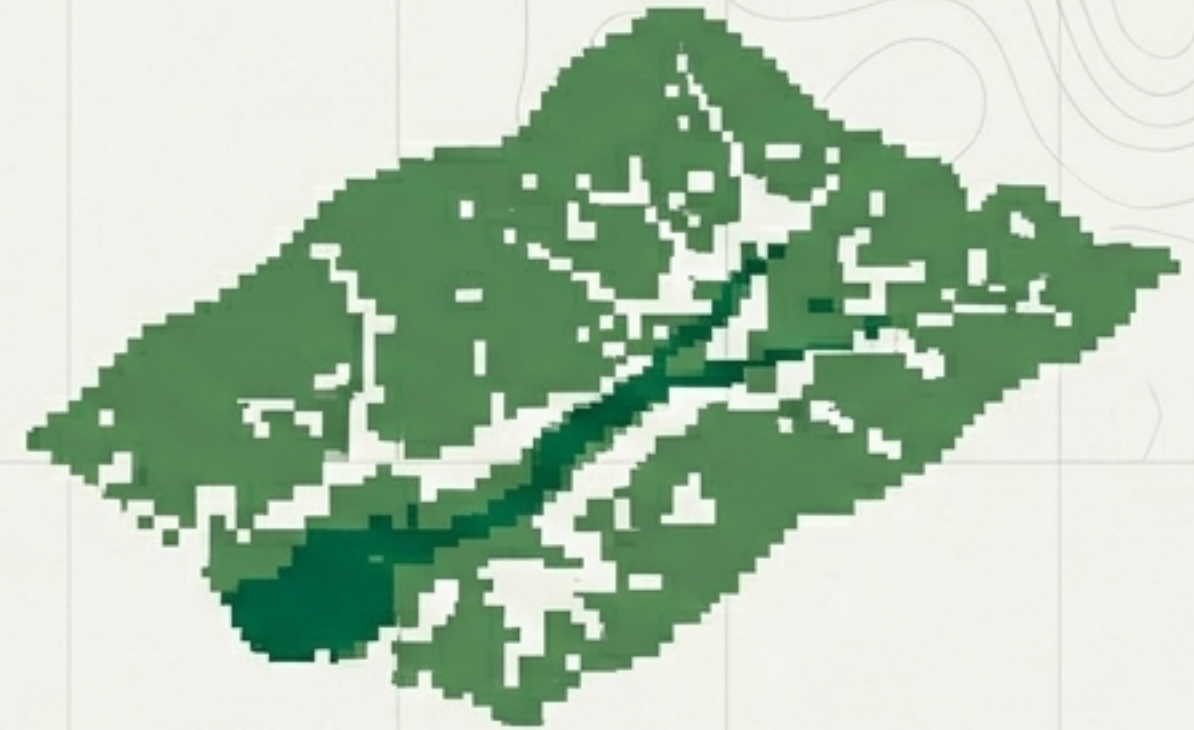
Steps 1 & 2: The Dual Lenses of Prediction and Reality

Step 1 — Physical Baseline (Nature)



- **Model:** SWAT / Mass-balance routing.
- **Inputs:** ERA5 (Precipitation), SRTM 30m (Elevation), FAO HWSD (Soil), ESA CCI (Land Cover).
- **Output:** $\hat{y}_{\text{physics}}$ (Predicted baseline downstream turbidity based purely on nature).

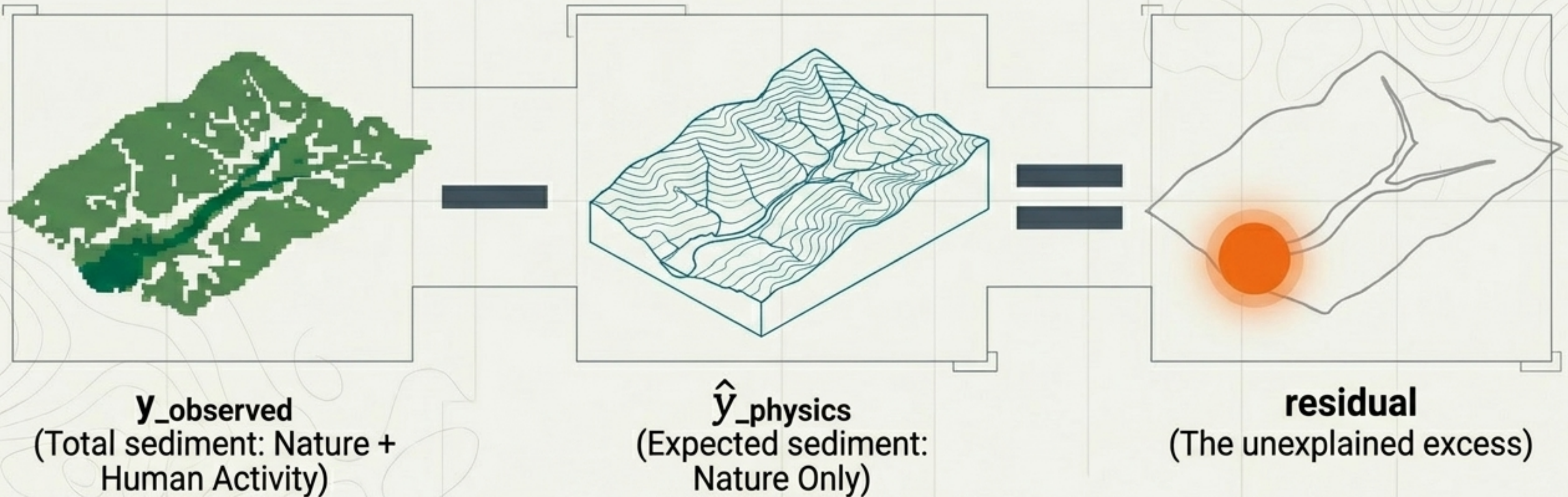
Step 2 — Satellite Observation (Reality)



- **Model:** Google Earth Engine / Sentinel-2 MSI (10m res, 2017-pres.).
- **Inputs:** Multispectral NDTI (Normalized Difference Turbidity Index) band ratios.
- **Output:** y_{observed} (Actual measured downstream turbidity).

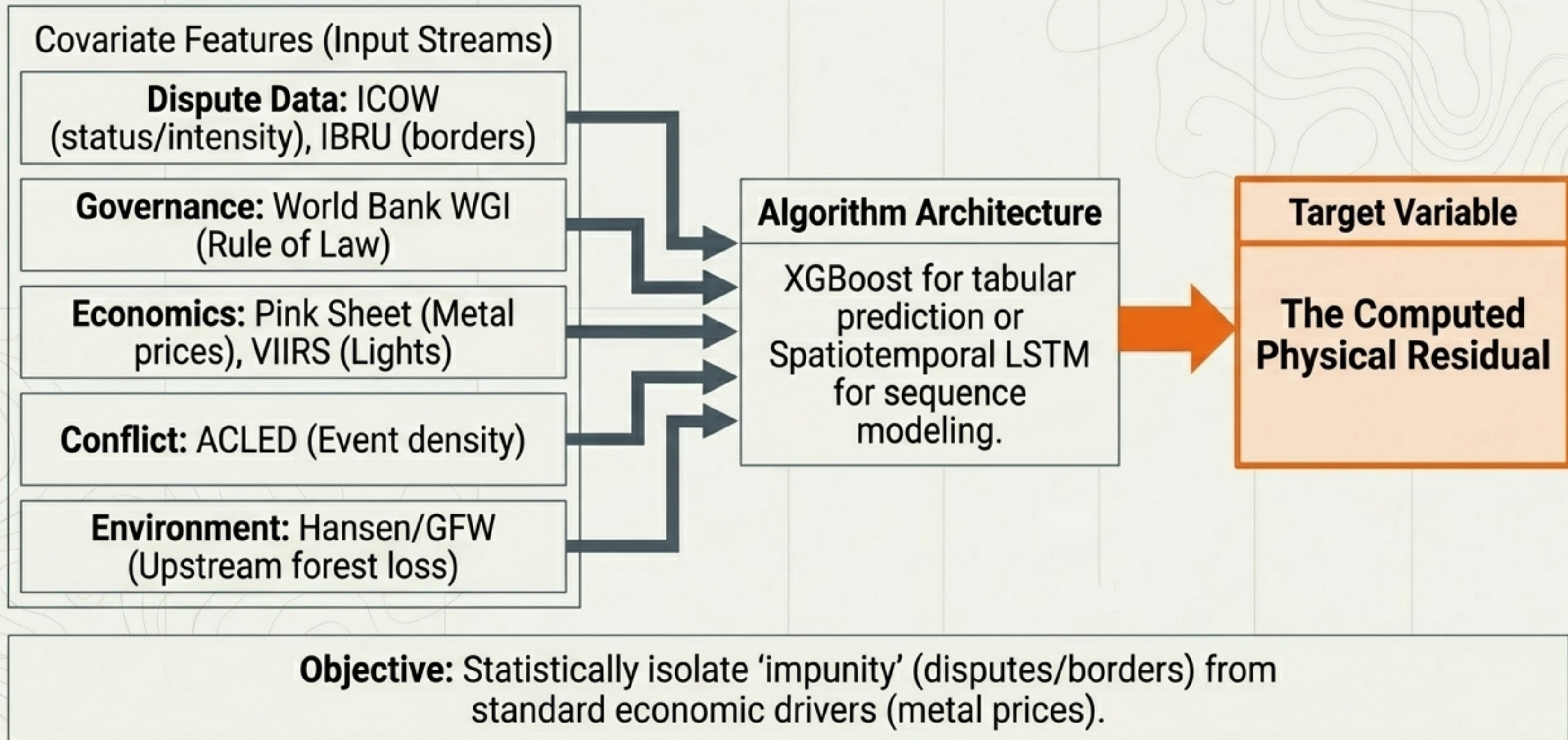
Step 3: Isolating the Geopolitical Residual

$$y_{\text{observed}}(t) - \hat{y}_{\text{physics}}(t) = \text{residual}(t)$$



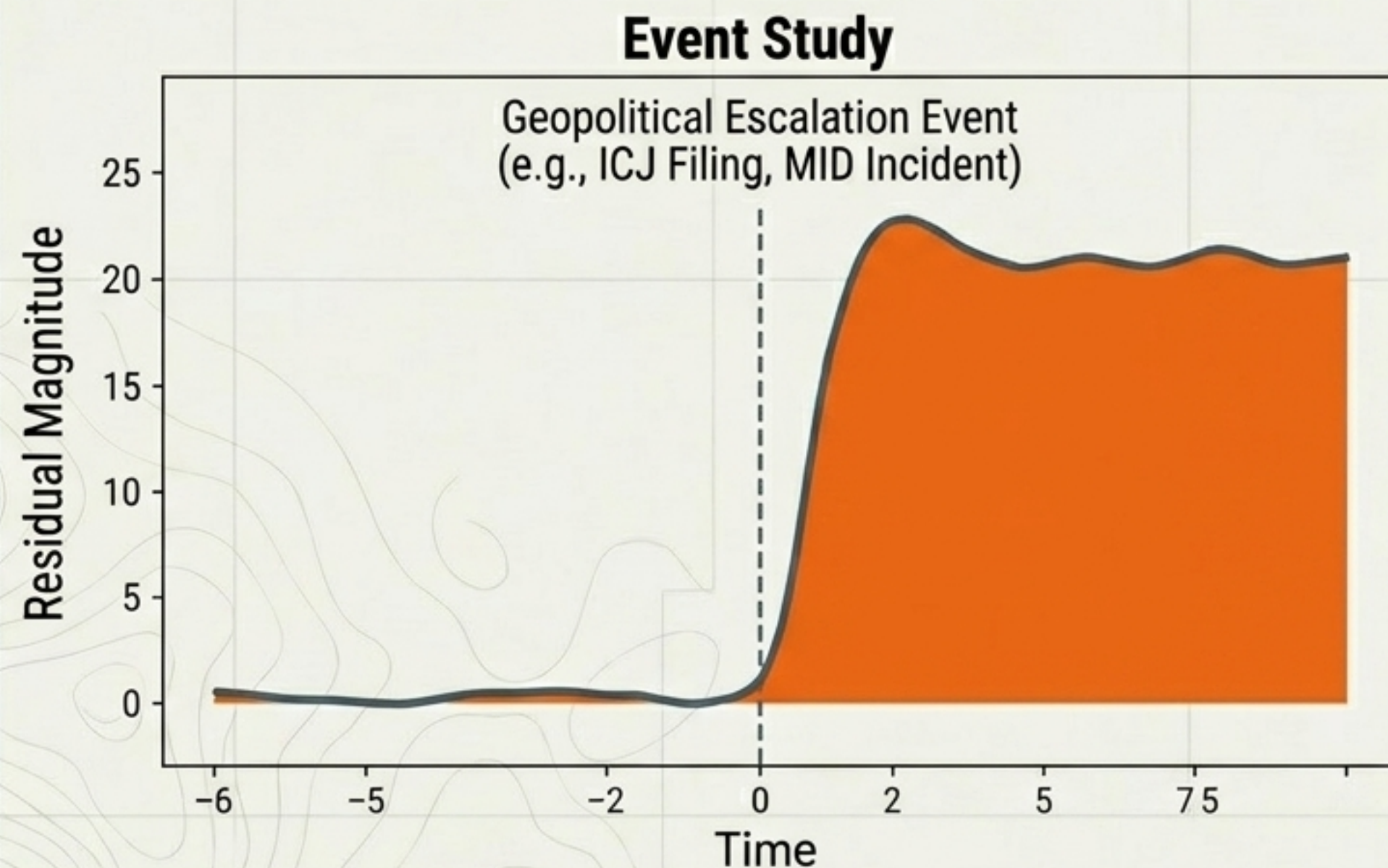
Key Insight: Positive residuals equate to excess sediment. This is the isolated fingerprint of extractive human activity that physical models cannot account for.

Step 4: The Machine Learning Predictor



Step 5: Causal Validation

Objective: Move beyond static correlation by testing how extractive anomalies respond to acute geopolitical shocks.



Methodology: Event study / Interrupted Time Series; Difference-in-Differences with Propensity Score Matching.

The Triggers: ICJ Filings (International Court of Justice), MID Dataset (Militarized Disputes).

The Hypothesis: An escalation in sovereignty disputes triggers an immediate, sustained spike in downstream turbidity residuals as domestic enforcement vanishes.

The Data & Tech Architecture

Layer 1: Physical & Observational (Free/Open)

ERA5 (Precipitation)	SRTM (DEM)	FAO HWSD (Soil)	ESA CCI (Land Cover)	Sentinel-2 MSI (Turbidity)	Hansen/GFW (Tree loss)	VIIRS (Lights)	GFW (Vessels)
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Layer 2: Geopolitical & Economic Predictors (Free/Academic)

ICOW (Disputes)	MID (Militarized events)	IBRU (Boundaries)	ACLED (Conflict events)	World Bank WGI (Governance)	World Bank Pink Sheet (Commodities)
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Layer 3: The Software Stack

- **Primary:** Python (rasterio, geopandas, xarray, scikit-learn, shap).
- **Spatial/Processing:** Google Earth Engine (GEE), QGIS, pysheds.
- **Causal/Spatial Modeling:** R (plm, spdep), PySAL.

100% of proposed datasets are strictly open-access or free for academic research.

Known Limitations & Strategic Mitigations

Hydrological Calibration

Challenge: Sparse stream gauge data in contested conflict zones limits exact SWAT calibration.

Mitigation: Deploy regionalized parameters and ungauged basin estimation techniques; utilize simplified mass-balance routing as a fallback.

Sensor Visibility

Challenge: Satellite turbidity indices only detect surface and alluvial mining, missing underground hard-rock extraction.

Mitigation: Explicitly bound the theoretical and empirical scope to surface-level extractive impacts.

Geographic Endogeneity

Challenge: Disputed zones may be inherently more mineral-rich, causing both the dispute and the mining.

Mitigation: Employ Propensity Score Matching (PSM) and Difference-in-Differences to compare geopolitically disputed zones with geologically identical uncontested zones.

Event Granularity

Challenge: Temporal granularity varies across dispute databases.

Mitigation: Cross-reference MID incident-level data with broader ICOW temporal bounds to ensure high-fidelity event studies.

The Bottom Line

The Shift in Paradigm

- Transitioning environmental geopolitics away from aggregate, static country-level statistics.
- Delivering dynamic, highly interpretable tracking at the precise resolution of individual river catchments.

The Ultimate Impact:

A scalable, automated, and mathematically rigorous framework to detect illegal resource extraction hiding inside geopolitical blind spots.